

AN ASYMPTOTIC DISTRIBUTION OF NONCENTRAL MULTIVARIATE DIRICHLET VARIATES

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1. Introduction

Let $A_r(p \times p)$ ($r=1, \dots, q$) and $B(p \times p)$ be independently distributed according to Wishart (Σ, m_r) and noncentral Wishart (Σ, n, Ω) respectively. Ω being the noncentrality parameter. Let

$$1.1 \quad L_r = E^{-1/2} A_r (E^{-1/2})',$$

$E^{1/2}$ being a lower triangular matrix such that $E^{1/2}(E^{1/2})' = E = \sum_{r=1}^q A_r + B$. The purpose of this paper is to derive an asymptotic distribution for $\prod_{r=1}^q |L_r|$. We shall first derive the joint density of $L_1, \dots, L_q$ which is called the noncentral multivariate Dirichlet density. Unfortunately this density cannot be written in an explicit form. From this density the h -th moment of $\prod_{r=1}^q |L_r|$ can however be derived.

In the last section the same is done for the complex case. A_r and B are then assumed to be complex variables, positive definite Hermitian, distributed according to complex Wishart (Σ, m_r) ($r=1, \dots, q$) and noncentral complex Wishart (Σ, n, Ω), respectively. Σ is also Hermitian.

2. Preliminary Results

Lemma 1. If R and S are two positive definite symmetric matrices of order p , then (Constantine (1963))

$$2.1 \quad \int_{E>0} \exp(-\frac{1}{2} \text{tr} RE) |E|^{a-1/2(p+1)} C_K(SE) dE \\ = \Gamma_p(a) |R|^{-a} {}_K C_K(R^{-1}S).$$

Lemma 2. If R is a positive definite symmetric matrix of order p , then (de Waal (1970))

$$2.2 \quad \int_{-1}^1 \int_0^1 \dots \int_0^1 \prod_{r=1}^q |L_r|^{a_r-1/2(p+1)} |I - \sum_{r=1}^q L_r|^{b-1/2(p+1)} \\ \cdot C_K(R \sum_{r=1}^q L_r) \prod_{r=1}^q dL_r \\ = (\Gamma_p(b) \prod_{r=1}^q \Gamma_p(a_r) / \Gamma_p(a+b)) (|a|_K / |a+b|_K) C_K(R), \text{ where } a = \sum_{r=1}^q a_r$$

From this lemma it also follows that

$$\begin{aligned}
 2.3 \quad & \int \cdots \int_{0 < L_r < I, 0 < \Sigma_r L_r < \Pi_r \frac{q}{I} |L_r|^{a_r - \frac{1}{2}(p+1)} |I - \Sigma_r L_r|^{b - \frac{1}{2}(p+1)}} \\
 & C_K(R(I - \Sigma_r L_r)) \Pi_r dL_r \\
 & = (\Gamma_p(b) \Pi_{r=1}^q \Gamma_p(a_r) / \Gamma_p(a+b)) / ((b)_K / (a+b)_K) C_K(R).
 \end{aligned}$$

3. The Joint Density of $L_1, \dots, L_q$

Theorem 1

If L_r is defined as in (1.1), then the joint density of $L_1, \dots, L_q$ is given by

$$\begin{aligned}
 3.1 \quad & (1/\Gamma_p(\frac{1}{2}n)) \Pi_{r=1}^q \Gamma_p(\frac{1}{2}m_r) / |2\Sigma|^{-\frac{1}{2}(m+n)} \exp(-\frac{1}{2}\text{tr}\Omega) \Pi_{r=1}^q |L_r|^{m_r - p - 1} \\
 & |I - \Sigma_r L_r|^{\frac{1}{2}(n-p-1)} \int_{E>0} e^{i\text{tr}(-\frac{1}{2}\Sigma^{-1}E)} |E|^{\frac{1}{2}(m+n-p-1)} \\
 & {}_0F_1(\frac{1}{2}n; \frac{1}{4}\Omega \Sigma^{-1} E^{\frac{1}{2}} (I - \Sigma_r L_r) (E^{\frac{1}{2}})') dE, L_r > 0, \Sigma_r L_r < I, m = \Sigma_{r=1}^q m_r.
 \end{aligned}$$

Proof

The joint density of $L_r (r=1, \dots, q)$ and E is easily obtained from the joint density of $A_r (r=1, \dots, q)$ and B . The density of $B>0$ (noncentral Wishart) is given by

$$(1/\Gamma_p(\frac{1}{2}n)) |2\Sigma|^{-\frac{1}{2}n} \exp(-\frac{1}{2}\text{tr}\Omega) \exp(-\frac{1}{2}\text{tr}\Sigma^{-1}B) |B|^{\frac{1}{2}(n-p-1)} {}_0F_1(\frac{1}{2}n; \frac{1}{4}\Omega \Sigma^{-1}B).$$

The jacobian being $J(A_1, \dots, A_q, B \rightarrow L_1, \dots, L_q, E) = |E|^{-\frac{1}{2}(p+1)q}$. Integration with respect to E gives the result. The density 3.1 is called the noncentral multivariate Dirichlet density.

From 3.1 we can obtain the moments of $\Pi_{r=1}^q |L_r|$ and $|I - \Sigma_r L_r|$ using 2.2 and 2.1 successively, i.e.

$$3.2 \quad E(\Pi_{r=1}^q |L_r|^{h_r}) = (\Gamma_p(\frac{1}{2}(m+n))) \Pi_{r=1}^q \Gamma_p(\frac{1}{2}m_r + h_r) / (\Gamma_p(\frac{1}{2}(m+n) + h))$$

$$\Pi_{r=1}^q \Gamma_p(\frac{1}{2}m_r)) \exp(-\frac{1}{2}\text{tr}\Omega) {}_1F_1(\frac{1}{2}(m+n); \frac{1}{2}(m+n) + h; \frac{1}{2}\Omega),$$

$$h = \Sigma_{r=1}^q h_r \text{ and}$$

$$3.3 \quad E(|I - \sum_{r=1}^q L_r|)^h = (\Gamma_p(\frac{1}{2}(m+n)) \Gamma_p(\frac{1}{2}n+h) / (\Gamma_p(\frac{1}{2}n) \Gamma_p(\frac{1}{2}(m+n)+h))) \\ \exp(-\frac{1}{2} \text{tr} \Omega) {}_2F_2(\frac{1}{2}n+h, \frac{1}{2}(m+n); \frac{1}{2}n, \frac{1}{2}(m+n)+h; \frac{1}{2}\Omega).$$

Since 3.3 is the same as equation 2.6 in de Waal (1968), the same asymptotic distribution (3.24) holds for $|I - \sum_r L_r|$.

4. Asymptotic Distribution for $\prod_{r=1}^q |L_r|$

An asymptotic distribution for $\prod_r |L_r|$ will now be derived using the moment 3.2.

Consider the function g of the form (de Waal (1968))

$$4.1 \quad g(t) = C(x^X / \prod_{r=1}^q x_r^{X_r})^{-2it} \rho \prod_{j=1}^p (\prod_{r=1}^q \Gamma(\rho x_r (1-2it) + \beta_r + \nu_j) / \\ \Gamma(\rho x (1-2it) + \beta + \nu_j + \gamma_j))$$

where C is a constant such that $g(0)=1$, ρ a chosen constant, $\beta_r = (1-\rho)x_r$, $\sum_{r=1}^q x_r = x$, $\sum_{r=1}^q \beta_r = \beta$ and $i^2 = (-1)$. Ignoring terms of order $\min(x_r^{-2})$ and higher, this function can be written as

$$4.2 \quad g(t) = (1-2it)^{-\frac{1}{2}f} (1 + \omega((1-2it)^{-1} - 1)) + O(x_{\min}^{-2}),$$

where $O(x_{\min}^{-2})$ means terms of order $\min(x_r^{-2})$ and higher.

$$4.3 \quad f = \sum_{j=1}^p ((q-1)(1-2\nu_j) + 2\gamma_j)$$

$$4.4 \quad \omega = ((1/2)\rho)(\sum_{j=1}^p (\sum_{r=1}^q (1/x_r) - 1/x) (\nu_j^2 - \nu_j + 1/6) - (\gamma_j^2 + 2\nu_j \gamma_j - \gamma_j)/x) \\ -(1-\rho)f).$$

The following theorem can now be proved:

Theorem 2

If $L_1, \dots, L_q$ are distributed according to 3.1, then

$$4.5 \quad P(Q - \rho \sum_{r=1}^q m_r \log |L_r| < z) = \sum_{k=0}^3 A_k P(\chi_{f_k}'(\text{tr } \Omega) < z) + O(m_{\min}^{-2})$$

where

$$Q = \rho p (\sum_{r=1}^q m_r \log(\frac{1}{2} m_r) - m \log(\frac{1}{2} m)),$$

$$\rho = 1 + ((n-p-1)/m - (\sum_{r=1}^q 1/m_r - 1/m) (2p^2 + 3p - 1)/6) / ((p+1)/(q-1) + 2n),$$

$$A_0 = 1,$$

$$A_1 = (1/\rho) / (tr \frac{1}{2} \Omega) / (1 - \rho + \frac{n}{m}),$$

$$A_2 = (1/\rho) / ((1/m) tr(\frac{1}{2} \Omega)^2 - (1 - \rho + \frac{n}{m}) tr \frac{1}{2} \Omega),$$

$$A_3 = (-1/\rho m) tr(\frac{1}{2} \Omega)^2,$$

and $\chi'^2_{f_k}(tr \Omega)$ is a noncentral chi-square variate with $f_k = \frac{1}{2} p(q-1)/(p+1) + np + 2k$

degrees of freedom and noncentrality parameter $tr \Omega$.

Proof

Let $W = ((\frac{1}{2}m)^{\frac{1}{2}mp} / \prod_{r=1}^q (\frac{1}{2}m_r)^{\frac{1}{2}m_r p}) \prod_{r=1}^q |L_r|^{\frac{1}{2}m_r}$ and $M = -2 \log W$. From 3.2 the characteristic function of ρM can be written as

$$4.6 \quad \theta(t) = E(W^{-2it\rho}) = ((\frac{1}{2}m)^{\frac{1}{2}mp} / \prod_{r=1}^q (\frac{1}{2}m_r)^{\frac{1}{2}p m_r})^{-2it\rho} E(\prod_{r=1}^q |L_r|^{-\frac{1}{2}(2it\rho)m_r})$$

$$= \exp(-\frac{1}{2} tr \Omega) \sum_K \sum_K [((\frac{1}{2}m)^{\frac{1}{2}mp} / \prod_{r=1}^q (\frac{1}{2}m_r)^{\frac{1}{2}p m_r})^{-2it\rho} \Gamma_p(\frac{1}{2}(m+n), K).$$

$$\prod_r \Gamma_p(\frac{1}{2}m_r(1-2it\rho)) / (\Gamma_p(\frac{1}{2}m(1-2it\rho) + \frac{1}{2}n, K) \prod_r \Gamma_p(\frac{1}{2}m_r)).] \cdot C_K(\frac{1}{2}\Omega) / k!$$

The term within square brackets is now of the form 4.1 with $x_r = \frac{1}{2}m_r$, $\nu_j = \frac{1}{2}(1-j)$, and $\gamma_j = \frac{1}{2}n + K_j$. Use was made of the expression $(a)_K = \Gamma_p(a, K) / \Gamma_p(a)$ where $\Gamma_p(a, K) = \pi^{\frac{1}{2}p(p-1)} \prod_{j=1}^p \Gamma(a + \frac{1}{2}(1-j) + K_j)$, $K = (K_1, \dots, K_p)$ being a partition of k and $\Gamma_p(a) = \Gamma_p(a, 0)$. Hence, from 4.2

$$4.7 \quad \theta(t) = \exp(-\frac{1}{2} tr \Omega) \sum_K \sum_K (1-2it)^{-\frac{1}{2} f_k} (1 + \omega_k((1-2it)^{-1} - 1)) C_K(\frac{1}{2}\Omega) / k! +$$

$$\alpha(m_{\min}^{-2})$$

with

$$f_k = \frac{1}{2}p(p+1)(q-1) + np + 2k \text{ and}$$

$$4.8 \quad \omega_k = (-1/2\rho)((1-\rho)f_k + n(p(n-p-1)/2 + 2k)/m + 2(\sum_j K_j^2 - \sum_j K_j j)/m - p(\sum_r (1/m_r) - 1/m)(2p^2 + 3p - 1)/12).$$

Choosing ρ such that $\omega_0 = 0$, we have ρ as given in the theorem. Let

$$a_1(K) = \sum_j K_j^2 - \sum_j K_j j,$$

then ω_k becomes

$$4.9 \quad \omega_k = (-1/\rho)((1-\rho + n/m)k + a_1(K)/m)$$

Substitute 4.9 in 4.7. The characteristic function of

$$\rho M = Q - \rho \sum_{r=1}^q m_r \log |L_r|,$$

Q being given in the theorem, becomes

$$4.10 \quad \theta(t) = \exp(-\frac{1}{2} \text{tr} \Omega) (1 - 2it)^{-\frac{1}{2} f_0} \sum_k \sum_K (1 + \omega_k((1 - 2it)^{-1} - 1)) \cdot C_K(\frac{1}{2}(1 - 2it)^{-1} \Omega / k! + O(m_{\min}^{-2})).$$

Using the results of Sugira and Fujikoshi (1969) for a symmetric matrix S

$$\sum_k \sum_K k C_K(S) / k! = \exp(\text{tr } S) \cdot (\text{tr } S)$$

$$\sum_k \sum_K a_1(K) C_K(S) / k! = \exp(\text{tr } S) \cdot \text{tr}(S)^2$$

and remembering that the characteristic function of a noncentral chi-square variate with f_k degrees of freedom and noncentrality parameter $\text{tr} \Omega$ is given by

$$\theta_k(t) = (1 - 2it)^{-\frac{1}{2} f_k} \exp((\text{tr} \Omega)it(1 - 2it)^{-1})$$

4.10 can be simplified to

$$4.1.1 \quad \theta(t) = A_0 \theta_0(t) + A_1 \theta_1(t) + A_2 \theta_2(t) + A_3 \theta_3(t) + O(m_{\min}^{-2})$$

A_i ($i = 0, 1, 2, 3$) being given in the theorem.

Using the inverse theorem (see Anderson (1958), p. 206), the theorem follows.

If $q=1$, theorem 2 in de Waal (1968) follows directly from this theorem in a much simpler form.

5. The Complex Case

In the complex case L_r ($r=1, \dots, q$) is defined as

$$L_r = E^{-1/2} \overline{A_r (E^{-1/2})'}, E^{1/2} \overline{(E^{1/2})'} = E,$$

where $E^{1/2}$ is a lower triangular matrix with real diagonal elements and complex nondiagonal elements. The joint density of $L_1, \dots, L_q$ can be derived in the same way as for the real case and is given by

$$5.1 \quad (1/\bar{\Gamma}_p(n)) \Pi_r \bar{\Gamma}_p(m_r) |\Sigma|^{-(m+n)} \exp(-\text{tr} \Omega) \Pi_r |L_r|^{m_r - p} |I - \Sigma_r L_r|^{n-p} \\ \int_{E=\bar{E}' > 0} \exp(-\text{tr} \Sigma^{-1} E) |E|^{(m+n-p)} {}_0\bar{F}_1(n; \Omega \Sigma^{-1} E^{1/2} (I - \Sigma_r L_r) \overline{E^{1/2}}') dE.$$

We only have to know that the density of B , complex noncentral Wishart (Σ, n, Ω) is given by James (1964)

$$5.2 \quad (1/\bar{\Gamma}_p(n)) |\Sigma|^{-n} \exp(-\text{tr} \Omega) \exp(-\text{tr} \Sigma^{-1} B) |B|^{n-p} {}_0\bar{F}_1(n; \Omega \Sigma^{-1} B), B > 0,$$

where ${}_0\bar{F}_1$ is the hypergeometric function with complex argument and defined as

$${}_0\bar{F}_1(a; S) = \Sigma_K \Sigma_K (1/[a]_K) \bar{C}_K(S)/k!,$$

$$[a]_K = \bar{\Gamma}_p(a, K) / \bar{\Gamma}_p(a) \text{ if the gamma function exists and}$$

$$\bar{\Gamma}_p(a, K) = \Pi_{j=1}^p \Gamma(a + 1 - j + K_j).$$

The jacobian $J(A_1, \dots, A_q, B \bullet L_1, \dots, L_q, E) = |E|^{pq}$

5.1 is called the complex noncentral multivariate Dirichlet density.

Let $\Omega=0$ and $q=1$ in 5.1. Integration with respect to E is now James (1964)

$$\int_{E=\bar{E}'} |E|^{(m+n-p)} \exp(-\text{tr} \Sigma^{-1} E) dE = \bar{\Gamma}_p(m+n) |\Sigma|^{m+n}.$$

The resulting density

$$5.3 \quad (\bar{\Gamma}_p(m+n)/\bar{\Gamma}_p(m)\bar{\Gamma}_p(n)) |L|^{m-p} |I-L|^{n-p}, \quad 0 < L < I,$$

is also the density of $L = \Sigma_r^{-1} L_r$ if B is distributed as a central complex Wishart (Σ, m) , $m = \Sigma_r m_r$. The following integral follows now from 5.1 if we let $\Omega=0$ and integrate with respect to $L_1, \dots, L_q$ over the space $\Sigma_r L_r = L$;

$$\begin{aligned} 5.4 \quad & \int \int_{\Sigma_r L_r = L} \Pi_r |L_r|^{m_r-p} |I - \Sigma_r L_r|^{n-p} \Pi_r dL_r \\ &= (\bar{\Gamma}_p(n) \Pi_r \bar{\Gamma}_p(m_r) / \bar{\Gamma}_p(m) \bar{\Gamma}_p(n)) |L|^{m-p} |I-L|^{n-p}. \end{aligned}$$

The following lemma can now be proved:

Lemma 3

If R is positive definite Hermitian of order p , then

$$\begin{aligned} 5.5 \quad & \int \int_{0 < L_r = \bar{L}_r < I, 0 < \Sigma_r L_r < I} \Pi_r |L_r|^{m_r-p} |I - \Sigma_r L_r|^{n-p} \bar{C}_K(R(I - \Sigma_r L_r)) \Pi_r dL_r \\ &= (\Pi_r \bar{\Gamma}_p(m_r) \bar{\Gamma}_p(n) / \bar{\Gamma}_p(m+n)) ([n]_K / [m+n]_K) \end{aligned}$$

Proof

Integrate 5.4 first with respect to $L_1, \dots, L_q$ over the space $\Sigma_r L_r = L$ using 5.4 and then with respect to L using

$$\begin{aligned} & \int_{0 < L = \bar{L}' < I} |L|^{m-p} |I-L|^{n-p} \bar{C}_K(R(I-L)) dL \\ &= (\bar{\Gamma}_p(m) \bar{\Gamma}_p(n) / \bar{\Gamma}_p(m+n)) ([n]_K / [m+n]_K) \bar{C}_K(R). \end{aligned}$$

This last integral can be checked from the complex noncentral multivariate F distribution (James (1964)).

From 5.1, applying lemma 3 and the integral (James (1964))

$$\int_{E=\bar{E}'>0} |E|^{m+n-p} \exp(-\text{tr} \Sigma^{-1} E) \bar{C}_K(RE) dE = \bar{\Gamma}_p(m+n) |\Sigma|^{m+n} \bar{C}_K(R\Sigma),$$

it follows that

$$5.6 \quad E(\Pi_r | L_r | h_r) = (\bar{\Gamma}_p(m+n) \Pi_r \bar{\Gamma}_p(m_r + h_r) / \bar{\Gamma}_p(m+n+h) \Pi_r \bar{\Gamma}_p(m_r)) \exp(-\text{tr} \Omega)$$

$${}_1\bar{F}_1(m+n, m+n+h; \Omega), h = \Sigma_r h_r$$

Applying the same procedure to 5.6 as in section 4, we obtain

$$5.7 \quad \omega_k = (-1/2\rho)((1-\rho)f_k + (n/m)(p(n-p)+2k) - (\Sigma_r(1/m_r)-1/m)p(2p^2-1)/6 + b_1(K)/m),$$

where

$$f_k = p^2(q-1) + 2np + 2k$$

and

$$b_1(K) = \Sigma_j K_j^2 - 2\Sigma_j K_j j + k.$$

If ρ is chosen such that $\omega_0=0$, ω_k becomes

$$5.8 \quad \omega_k = (-1/\rho)((1-\rho+n/m)k + b_1(K)/m)$$

If we substitute 5.8 in the characteristic function

$$\theta(t) = \exp(-\text{tr} \Omega) (1-2it)^{-1/2} f_0 \Sigma_k \Sigma_K (1+\omega_k((1-2it)^{-1}-1)) \bar{C}_K((1-2it)^{-1} \Omega)/k! + O(m_{\min}^{-2})$$

of ρM , we need the results (S being Hermitian)

$$5.9 \quad \Sigma_k \Sigma_K k \bar{C}_K(S)/k! = \text{tr}(S) \text{etr}(S)$$

$$5.10 \quad \Sigma_k \Sigma_K b_1(K) \bar{C}_K(S)/k! = \text{tr}(S)^2 \text{etr}(S)$$

5.9 follows easy. To prove 5.10 we apply the procedure of Sugiura and Fujikoshi (1969) using the fact that

$$5.11 [a]_K = a^k(1 - b_1(K)/2a + O(a^{-2})).$$

We can now state the theorem:

Theorem 3

If the complex variables $L_1, L_2, \dots, L_q$ are distributed according to 5.1, then

$$5.12 P(Q - 2\rho p \Sigma_r m_r \log |L_r| < z) = \Sigma_{k=0}^3 A_k P(\chi_{f_k}^{\prime 2}(\text{tr} 2\Omega) < z) + O(m_{\min}^{-2})$$

where

$$Q = 2\rho p(\Sigma_r m_r \log m_r - m \log m),$$

$$f_k = p^2(q-1) + 2np + 2k,$$

$$\rho = 1 + (n(n-p)/m - (\Sigma_r 1/m_r) - 1/m) \cdot (2p^2 - 1)/6 / (p(q-1) + 2n),$$

$$A_0 = 1,$$

$$A_1 = (1/\rho)(1 - \rho + n/m)\text{tr}\Omega,$$

$$A_2 = (1/\rho)((1/2m)\text{tr}\Omega^2 - (1 - \rho + n/m)\text{tr}\Omega),$$

$$A_3 = (-1/2\rho m)\text{tr}\Omega^2$$

and

$\chi_{f_k}^{\prime 2}(\text{tr} 2\Omega)$ a noncentral chi-square variate with f_k degrees of freedom and noncentrality parameter $\text{tr} 2\Omega$.

REFERENCES

- ANDERSON, T.W. (1958). *An introduction to multivariate statistical analysis*. Wiley, N.Y.
- CONSTANTINE, A.G. (1963). Some noncentral distribution problems in multivariate analysis. *Ann. Math. Statist.* **34**, 1270-1285.

- DE WAAL, D.J. (1968). An asymptotic distribution for the determinant of a noncentral B statistic in multivariate analysis. *S. Afr. Statist. J.* **2**, 77-84.
- DE WAAL, D.J. (1968). *Nie-sentrale meerveranderlike Beta verdelings*. Unpublished Ph.D. thesis, University of Cape Town.
- DE WAAL, D.J. (1970). Distributions connected with a multivariate Beta statistic. *Ann. Math. Statist.* **41**, 1091-1095.
- JAMES, A.T. (1964). Distribution of matrix variates and latent roots derived from normal samples. *Ann. Math. Statist.* **35**, 475-501.
- SUGUIRA, N. and FUJIKOSHI, Y. (1969). Asymptotic expansions of the non-null distributions of the likelihood ratio criteria for multivariate linear hypotheses and independence. *Ann. Math. Statist.* **40**, 942-952.

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